3264 CONICS

TOYESH JAYASWAL

A heavy warning used to be given that pictures are not rigorous; this has never had its bluff called and has permanently frightened its victims into playing for safety

— John Littlewood

1. Introduction

This talk is about an area of math called enumerative geometry. As the name suggests, enumerative geometry deals with counting how many geometric objects satisfy a certain amount of constraints. Often times in such problems, the actual numerical answer is not particularly interesting, but the fact that an answer is computable at all, and the methods used to compute these numbers, is where all the beauty lies. Often, these problems are solved by studying associated "moduli spaces" or "classifying spaces" i.e. parameter spaces. The geometry of these classifying spaces can give us answers to geometric problems in the space we care about. This talk will discuss the following surprising result:

> **Theorem 1.1.** Given 5 conic sections in "general position" on the plane, there are exactly 3264 conic sections which are simultaneously tangent to all 5 of the conics.

Here, "general position" is a common invocation in enumerative geometry which says that if the contraints are "too special", the number may be wrong, but such constraints can be perturbed slightly to yeild the "correct" (i.e. for random enough constraints) answer. For example, 3 points usually define a triangle, unless they are collinear or some of the points overlap. But in either bad scenario, we can move the points a little bit to make them define a non-degenerate triangle. The reason that there are 5 conic sections as constraints will be seen later to be the right number to induce a finite answer and thus have an interesting counting problem.

If this material seems interesting, look more into enumerative geometry! There's a lot of beautiful results pertaining to Grassmannians to solve problems regarding sub-planes (e.g. given 4 lines in $\mathbb{R}^3$, there are generically exactly 2 lines passing through all 4 of them), line bundles (used to prove the very famous 27 lines on a cubic surface), and much more! A more advanced book on the topic is "3264 and all that" by Eisenbud and Harris.

2. 1 Conic Through 5 Lines - An Introduction to Counting Geometrically

Before tackling our big problem, we will first explore a simpler problem which outlines the fundamental principles of enumerative geometry problems. The first subproblem that we will solve is:

> **Theorem 2.1.** Through any 5 points on the plane (in general position), there is a unique conic section that passes through all of them.

The method to proving this will be quite straightforward - essentially we just write out the equation of the most general conic section and then impose the constraints of passing through points ans solve for coefficients. However we will phrase things in a very geometric manner and introduce our main character throughout this talk: the parameter space of conics in the plane. We notice that a conic section can be expressed by an equation with six coefficients:

$$a_0x^2 + a_1y^2 + a_2xy + a_3x + a_4y + a_5 = 0.$$

However, because rescaling all our coefficients by the same amount does not change the resulting locus of points (x, y) which satisfies the equation, the space of conic sections is actually these six coefficients *up to scaling*. In other words, a conic section is defined by the tuple $(a_0, a_1, a_2, a_3, a_4, a_5)$, but we have $(a_0, a_1, a_2, a_3, a_4, a_5) \sim (\lambda a_0, \lambda a_1, \lambda a_2, \lambda a_3, \lambda a_4, \lambda a_5)$ for all $\lambda \in \mathbb{R}$. You may recognize this space as $\mathbb{P}^5$, the 5 dimensional projective space. Note that since we have a 5 dimensional parameter space, we expect that if we impose 5 constraints on our conics, we should have a finite number of conics that satisfies all of them, and thus we have a well-defined counting problem (any less constraints and we expect an infinite number of solutions; any more and we expect none).

(Note: the keen of you may have noticed that not all choices of coefficients work - we may end up with $1 = 0$ as a "conic" for example. In order to fix this problem, we should really be working in projective space, $\mathbb{RP}^2$ instead of $\mathbb{R}^2$. In fact, we should *really* be working in $\mathbb{CP}^2$ and do everything complex projectively for everything to work correctly. For example, a conic and a line should intersect at two points, but if we change the line and conic, the points of intersection may escape off to infinity, or off to the complex numbers. Thus, I will usually use $\mathbb{P}$ for projective space and be ambiguous on whether I mean over $\mathbb{R}$, for intuition, or $\mathbb{C}$, for accuracy.)

Say we have some point (u_1, v_1), how many conics pass through this point? Plugging this into our equation for a general conic, we get

$$u_1^2 a_0 + u_2^2 a_1 + u_1 u_2 a_2 + u_1 a_3 + u_2 a_4 + a_5 = 0,$$

or in other words, a linear constraint on our six coefficients. The solution set to a linear equation in $\mathbb{P}^5$ is a hyperplane i.e. a copy of $\mathbb{P}^4$ (a linear equation in the plane defines a line; a linear equation in 3 dimensional space defines a plane). If we impose that our conic passes through another point (u_2, v_2), it must lie on the two different hyperplanes corresponding to the linear constraints from each equation. If we have 5 points, each point induces a linear constraint on our conic, and a conic passes through all 5 points iff it satisfies all 5 constraints. Geometrically, this is asking how many points in $\mathbb{P}^5$ simultaneously lie on 5 different hyperplanes $\mathbb{P}^4$. This follows from the general statement that n hyperplanes in n dimensional space will intersect at a single point (think of the coordinate hyperplanes).

Voila! We have proven that there is only one conic that passes through any 5 points. And we didn't even have to solve a single equation! We have now already seen some of the general principles of enumerative goemetry problems:

- Study the parameter space of the type of geometric object you're trying to study
- The space of objects that satisfy a single constraint will form a codimension 1 (dimension 1 less than the ambient dimension) subspace of your parameter space
- In order to expect a finite number of solutions, the number of constraints should equal the dimensionality of the parameter space - the intersection of a codimension n and codimension m space will generically be codimension $m + n$
- Try to use the geometry of your parameter space to find the intersections of your hypersurfaces

3. Mixing up Points and Lines

In this section, we will introduce a new constraint on our conics: instead of constraining the conic to pass through a point, we will constrain the conic to be tangent to some given line. Then we can ask questions like "how many conics are tangent to five given lines?" or mix up the conditions such as "how many conics pass through three given points and are tangent to two given lines?" (Note the reason to demand tangency instead of just intersecting a line is because (at least over $\mathbb{C}$), any conic intersects any line, so this is not really a constraint).

We start by analyzing when a conic is tangent to a line. Note that we can parametrize a line as the set of points $(t, ut + v)$ for a parameter t that varies in $\mathbb{R}$ and fixed values (u, v). We can substitute $x = t$ and

$y = ut + v$ into the function $F_a(x, y) = a_0x^2 + a_1y^2 + a_2xy + a_3x + a_4y + a_5$ to obtain a function

$$F_a(t) = F_a(x = t, y = ut + v) = f_2(a_i)t^2 + f_1(a_i)t + f_0(a_i)$$

(expand binomials and collect linear and quadratic terms in t) where f_1, f_2, f_3 are all linear functions of the a_i. The reason to do this is as follows: $F_a(x, y) = 0$ precisely the equation of our conic section defined by the tuple $a = (a_0, a_1, a_2, a_3, a_4, a_5)$. Thus, the values of t when $F_a(t) = 0$ precisely correspond to the values of t where the line intersects the conic. Note that since F_a is a quadratic in t, we expect that a line intersects a conic generically (and at most) at two points, and this is true. We can imagine that as we move a line to become tangent to a conic, these two points of intersection will coincide i.e. $F_a(t)$ will have a double root. Now, a quadratic equation has a double root precisely when the discriminant of its coefficients vanish. Thus, for a fixed line, a conic is tangent to our line precisely when $f_1(a_i)^2 - 4f_0(a_i)f_2(a_i) = 0$. Note that since f_1, f_2, f_3 are all homogenous of degree 1, the equation $f_1(a_i)^2 - 4f_0(a_i)f_2(a_i) = 0$ defines a homogenous degree 2 equation on our parameter space $\mathbb{P}^5$, thus defining a "degree 2 hypersurface" (similar to a conic section in the plane, or a sphere/paraboloid/hyperboloid in 3 dimensional space). This equation will depend on the parameters u, v which define our line. It's an exercise to compute it explicitly (and test it out!).

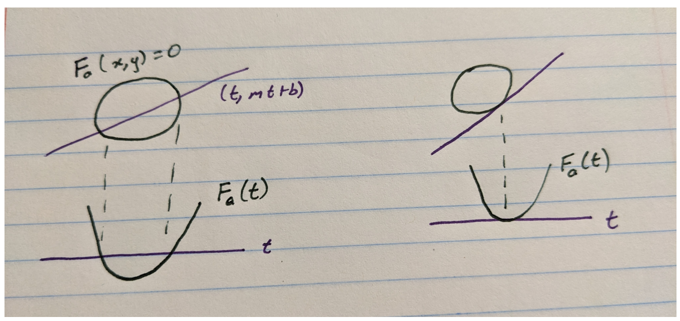


FIGURE 1. The equation $F_a(t) = 0$ determines the intersection of the conic with a line and can be used to determine tangency.

Let's now ask the following question: how many conic sections pass through 4 given points and are tangent to a given line. Translating this question into our parameter space, we've seen that the four points give us four hyperplanes in $\mathbb{P}^5$ and the line gives us a constraint hypersurface of degree 2. Now, we need to figure out how many points in the parameter space simultaneously lie on all 5 hypersurfaces. To solve this, we must explore the very powerful

3.1. **Bézout's Theorem.** Bézout's theorem gives us a very efficient way to compute the intersection of hypersurfaces defined as the zero sets of homogenous polynomial equations.

> THEOREM 3.1 (Bézout). Given n hypersurfaces (in general position) in $\mathbb{P}^n$ of degree $d_1, \cdots, d_n$ (the degree of a hypersurface is the degree of the polynomial used to define it) respectively, there are exactly $d_1d_2\cdots d_n$ many points that simultaneously lie on all of them.

It should be mentioned here that for the theorem to work, we actually need to be working over an algebraically closed field like $\mathbb{C}$, but in nice enough conditions, we'll be able to see everything over $\mathbb{R}$. In the plane, this says that a degree m equation and a degree n equation will intersect at mn many points. For example, a circle and a line intersect at two points. In 3 dimensional space, two planes and a sphere intersect at $1 \times 1 \times 2 = 2$ points (two planes intersect in a line which intersects a sphere at 2 points). You can convince yourself that if you take three spheroids $(\left(\frac{x}{a}\right)^2 + \left(\frac{y}{b}\right)^2 + \left(\frac{z}{c}\right)^2 = 1)$, they should intersect at $2 \times 2 \times 2 = 8$ points.

In order to see the last example, it may help to think about "very flattened spheroids". By thinking about these, you may already roughly see what the main idea is: in the plane, for example, a degree m polynomial can be made to look like a union of m lines while a degree n polynomial can be made to look like a union of n lines, and these will intersect at mn many points. Modifying this idea a little, we can actually get a sketch for the full theorem. Let $F_1(x) = (x_1)(x_1 - 1)\cdots(x_1 - d_1)$ (a union of d_1 hyperplanes parallel to a coordinate hyperplane) and similarly $F_k(x) = (x_1)(x_1 - 1)\cdots(x_1 - d_k)$. For these polynomials, it is not too difficult to see that the total number of points of intersection is going to be $d_1 d_2 \cdots d_n$ (any n planes intersect at exactly 1 point). Now, given any n polynomials $G_i(x)$ of degree d_i respectively, we can "interpolate" from $G_i(x) = 0$ to $F_i(x) = 0$ via the functions $\tilde{F}_i(t, x) = G_i(x)(1 - t) + F_i(x)t = 0$ (for every t, $\tilde{F}_i(t, x) = 0$ defines some hypersurface which starts at $G_i(x) = 0$ at $t = 0$ and ends at $F_i(x) = 0$ at $t = 1$). The final picture we have is the union of hyperplanes which we saw to have $d_1 d_2 \cdots d_n$ many points while the first picture is the problem we're trying to solve. Now, as t varies a little bit, the hypersurfaces don't change much and so the number of points of intersection remains the same. The technical point is that in fact (if counted "correctly"), the number of points of intersection doesn't change regardless of the value of t, and in particular it is the same when $t = 0$ and when $t = 1$. (A good question: why does the degree of G_i have to be the same as the degree of F_i - hint: think projectivization).

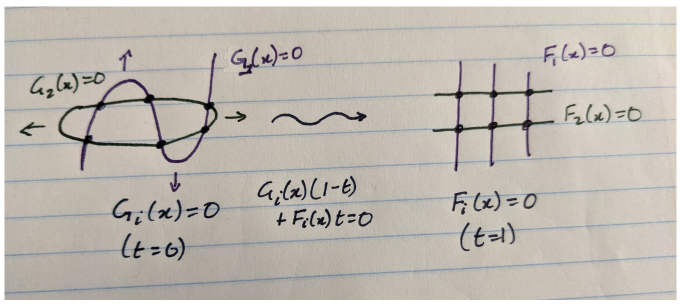


Figure 2. An example of Bézout's theorem, and the deformations used to prove it.

Now back to our main problem: how many conics pass through four points and are tangent to a given line i.e. how many points in $\mathbb{P}^5$ simultaneously lie on four hyperplanes and a degree two hypersurface? Bézout gives us a very simple answer for this! It's $1^4 \times 2 = 2$. Now how about a conic that passes through three points and two lines? Well that should be the number of points that pass through 3 hyperplanes and 2 degree two hypersurfaces. Thus, we have $1^3 \times 2^2 = 4$ solutions. How about 2 points and 3 lines? The answer should be 8! Right? Well... not quite.

4. Double Lines and the Need for Duality

Consider the following "conic section": $(x-1)^2 = 0$. It's not *really* a conic section. It's just a line - well, a line counted twice, but certainly not something we want to count towards our final solution. Unfortunately, it causes enough problems to trick our algebraic methods into thinking that it is a valid solution. The coefficients satisfy the constraint of passing through any point that that the line passes through. The more important issue, however, is that any intersection of the "double line conic" with any other line will be counted as a point of tangency! (Convince yourself that the corresponding discriminant will vanish.)

In fact, any two lines must intersect, so this causes a problem for any line we choose. Consider the problem of a conic that passes through two points and is tangent to three lines. We can just choose the line that passes through the two points, and since it intersects the other three lines and "is tangent to them", it will pass all of our constraints. Thus, we want to be careful to exclude these impostor conics. The reason this problem did not arise before is because we had at least 3 points, and if the 3 points are not collinear (which they generically won't be), we can't have an impostor line pass through all three.

There is a very elegant solution to this problem, which is the principle of duality. In the projective geometry talk, we saw that we can think of $\mathbb{RP}^2$ as being the space of lines in $\mathbb{R}^3$ through the origin while lines in the projective plane correspond to planes passing through the origin. This sets up a 1-1 correspondence between points in $\mathbb{P}^2$ and lines in $\mathbb{P}^2$. This correspondence is called duality. More precisely, we have a copy of $\mathbb{P}^2$, which I'll call $\mathbb{P}^{2*}$ where every point in this "dual copy" $\mathbb{P}^{2*}$ corresponds to the line in $\mathbb{R}^3$ orthogonal to the plane representing some line in our original $\mathbb{P}^2$. Now, if we take a nonsingular conic section, we can construct a "dual curve" as follows: for a point p on the conic, take the line that is tangent to the conic at that point and then take the dual point to this tangent line (which lives in $\mathbb{P}^{2*}$). As we move p around our conic, we get a curve traced out by our dual points in $\mathbb{P}^{2*}$. It turns out that this dual curve will also be defined by some polynomial equation. In fact, it will also be a conic!

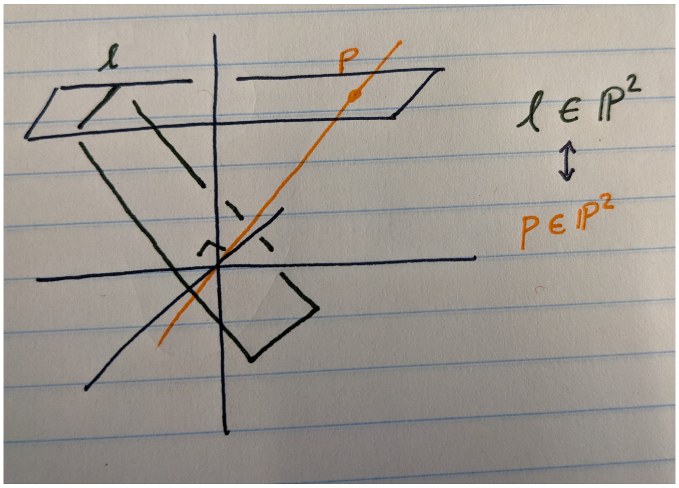


FIGURE 3. There is a 1-1 correspondence between lines through the origin in $\mathbb{R}^3$ and orthogonal planes through the origin.

The reason it's a conic is as follows: a line in $\mathbb{P}^{2*}$ corresponds to a single point in the original $\mathbb{P}^2$. The reason is the points on this line all correspond to lines in the original $\mathbb{P}^2$ which all intersect at one point (in $\mathbb{R}^3$, this means if you take a plane (representing a line on $\mathbb{P}^{2*}$), the line perpendicular to this plane (representing a point on $\mathbb{P}^2$) lies in every plane orthogonal to a line in the original plane). This is why its called duality! The process of taking duals also swaps incidence with intersection - a curve contains a point iff the dual curve intersects the dual line. Thus the degree of the dual curve is the number of times it intersects a generic line (by Bézout) which can be interpreted as the following geometric condition on our original curve: take a generic point (which is dual to a generic line), how many lines tangent to our conic (points on the dual curve) also pass through this point? Looking at a circle first, it should not be too difficult to convince yourself that the answer is 2. Thus, the dual to a conic section is another conic section.

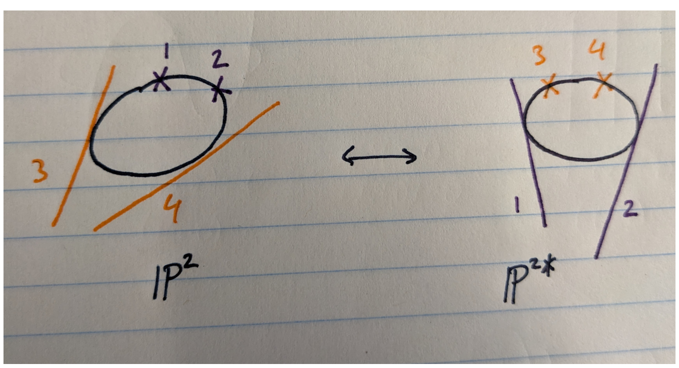


FIGURE 4. Duality swaps tangent lines on one conic with points on the dual conic.

Thus we have a 1-1 correspondence between (nonsingular) conic sections on $\mathbb{P}^2$ and (nonsingular) conic sections on $\mathbb{P}^{2*}$. The description for the construction of the dual curve doesn't really work for singular conics (two intersecting lines, or a double line). We can thus dualize our problem and ask for the associated problem on $\mathbb{P}^{2*}$. For instance, "how many conics pass through 2 points and tangent to 3 lines on $\mathbb{P}^2$" is the same question as "how many conics are tangent to 2 lines and pass through 3 points on $\mathbb{P}^{2*}$" (at least on nonsingular conics). The answer to the latter problem is 4, and so the answer to the former is also 4! We can do the rest of our problems like this as well, e.g. there is 1 conic (in $\mathbb{P}^2$) tangent to 5 lines since there is 1 conic (in $\mathbb{P}^{2*}$) that passes through 5 points.

To summarize,

Points	0	1	2	3	4	5
Lines	5	4	3	2	1	0
Conics	1	2	4	4	2	1

Table 1. Table of values for lines, points, and conics

5. 3264 Conics

Now that we've finished our warmups, we proceed to our main problem: how many conic sections are tangent to 5 given conic sections. First, we try and find the degree of the hypersurface in our classifying space $\mathbb{P}^5$ which describes which conics are tangent to a given conic. In order to do this, we proceed in a similar manner to finding which conics are tangent to a given line.

We begin by discussing a very clever parametrization of a conic section. The construction is as follows: fix a point p on the conic section. Take the set of lines through the point p with slope s (the vertical line will have slope $\infty \in \mathbb{P}^1$). Since a line intersects a conic at 2 points, and p already lines both on the line and the conic, any such line must intersect the conic at exactly one other line q_s (for the slope s that gives the unique line tangent to the conic at p, let $q_s = p$). Conversely, given q_s, s can be found by the line connecting p and q_s. This gives a 1-1 correspondence between the points of the line $\mathbb{P}^1$ (given by the slopes s) and the points of the conic (q_s), which parameterizes it in terms of rational functions. For example, for the circle with $p = (-1, 0)$, it can be checked that this parametrization is $\left(\frac{1-s^2}{1+s^2}, \frac{-2s}{1+s^2}\right)$, or, more nicely in projective coordinates, $[1 - s^2 : -2s : 1 + s^2]$ (fun fact: this allows you to generate all pythagorean triples, over any field!).

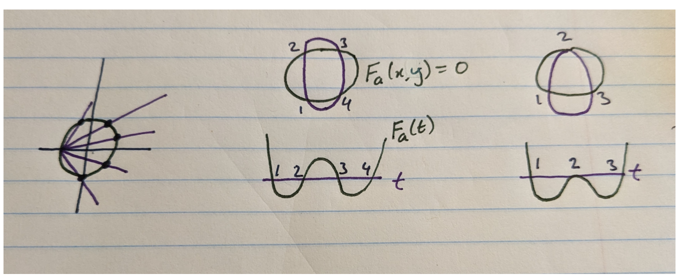


Figure 5. Parametrization of the circle with lines (left), $F_a(t)$ detecting intersections with the conic (middle), $F_a(t)$ detecting tangency with the conic (right).

If we now take the equation of our conic $F_a(x, y) = 0$ and substitute our parametrization $(x(s), y(s))$ of some fixed conic C, we get (after clearing denominators, or alternatively working projectively) a degree 4 equation (degree 2 parametrization $\times$ degree 2 equation F) $G(s) = 0$ whose zeros correspond to when the conic intersects C. To test if any of these are points of tangencies, we check again if $G(s)$ has any double root. Note that a polynomial $G(s)$ has a double root at s_0 iff $G(s_0) = G'(s_0) = 0$, so that to check if there is a double root (i.e. at least one point of tangency), we need to check if $G(s)$ and $G'(s)$ have a common root. There is a very elegant object called the resultant of two polynomials f, g which is a polynomial in the coefficients of f and g and vanishes iff f and g share a common root. Taking the resultant of a polynomial with its derivative (and dividing by the coefficient of x^n) gives a polynomial called the discriminant which vanishes iff the polynomial has a double root. The discriminant of a degree n polynomial has degree $2n - 2$, and thus we see that knowing when a conic $F_a(x, y) = 0$ is tangent to a given conic C is a degree $2 \times 4 - 2 = 6$ degree equation on the coefficients a_i (given by the discriminant of $G(s)$).

Thus, if we have 5 conic sections and demand our conic to be simultaneously tangent to all 5, by Bézout, we expect there to be $6^5 = 7776$ conics that satisfy our problem. This was the answer first proposed by Steiner when he posed this problem. However, it is incorrect! The reason is again because of the pesky double lines. Any line intersects any conic, and again our algebraic methods think that these double lines are all tangent. So in fact, any double line works, and there are infinitely many problems! Bézout's theorem only works if the hypersurfaces intersect "generically", that is they aren't tangent to one another, and don't contain any common connected components. However, every degree 6 hypersurface contains the locus of all points in $\mathbb{P}^5$ which represent double lines. But not all hope is lost! We will now show how you can "remove" this bad intersection of the hypersurfaces, but it will require a more detailed understanding of this subset of double lines.

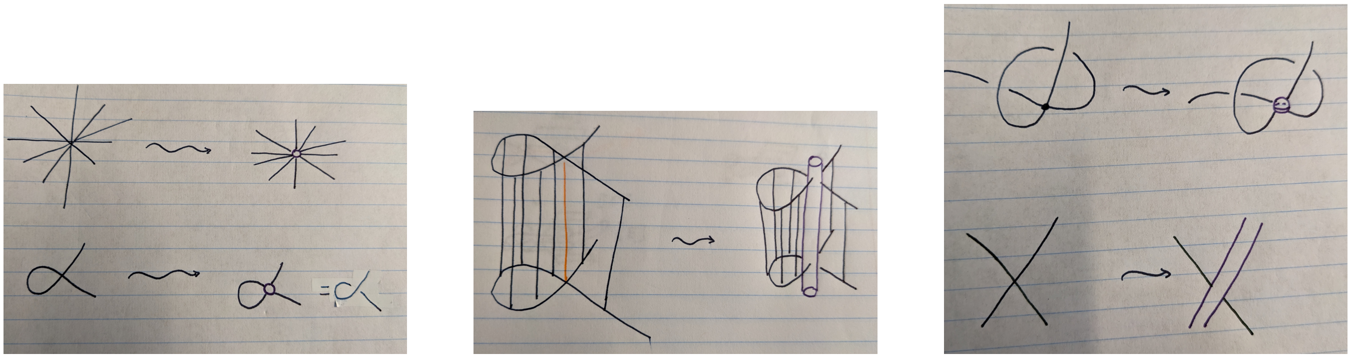
5.1. **The Space of Double Lines.** We now describe in a bit more detail what the geometry of the space of the set of double lines in $\mathbb{P}^5$ looks like. Note that all double lines are of the form $(b_0x + b_1y + b_2)^2 = 0$. Thus, similar to the space of conics, a double line is specified by a triple of numbers (b_0, b_1, b_2), but only up to scaling. So what we have is a copy of $\mathbb{P}^2$ parametrized by the homogeneous coordinates $[b_0 : b_1 : b_2]$ in $\mathbb{P}^5$. Explicitly, we can expand out the square to write the a_i as a parametric function of b_i e.g. $a_0 = b_0^2$ (coefficient of x^2) and $a_2 = b_0b_1$ (coefficient of xy). Thus, the set of double lines forms a two dimensional subspace of $\mathbb{P}^5$. Now, any degree 6 hypersurface consisting of conics which describes the conics tangent to a given conic will contain this two dimensional subspace. The thing to notice (or hope for) is that the hypersurfaces "approach this two dimensional space from different directions". We will now describe how keeping track of these different directions can allow us to "pull apart this bad intersection".

5.2. **Blowups.** Blowups are a beautiful construction in algebraic geometry which allows you to separate two things which intersect based on the direction that they approach the intersection. A classical application of this technique is the resolution of singularities of curves. The curve $y^2 = x^3 - x^2$ near the origin looks like the union of the lines $y = x$ and $y = -x$ i.e. it intersects itself. It is nicer to have curves that do not have such self intersections since they cause problems, for example, with doing calculus.

One thing to notice is that these two branches approach the origin from different directions. A clever way that we can separate these intersections is to replace the origins with a copy of $\mathbb{P}^1$ which represents the different directions a curve can hit the origin. To visualize this, you should remove the origin, glue in a small ("infinitesimal") circle and glue opposite ends of the circle. This is the model of projective space that sees it as a sphere with antipodal points identified. The way I like thinking about the curve at this new blown up point is it enters the circle from one side which is a "portal" to the other side where it continues. The curve now intersects this circle at two different points so we've separated the self intersection.

We can consider the same equation in $\mathbb{R}^3$, and now it looks like a sheet which self intersects along the $z-$axis. We can do a similar thing where instead of putting a circle around the origin, we put a small cylinder around the $z-$axis and identify antipodal points to form a "portal" through the cylinder. This separates the intersection of the sheets. Instead, if we have e.g. a curve in $\mathbb{R}^3$ which self intersects at the origin, we can remedy the problem by blowing up the origin into a sphere and identifying opposite points to get a copy of $\mathbb{RP}^2$ at the origin. The general procedure to blowup a subspace is to "thicken" the subspace into a tube, remove the inside of the tube, and identify opposite points of the tube to form a "portal". We can, for example, take a circle and blow it up to get a torus, and then identify opposite points of each circle around every point.

Note that if we take a plane, the thickening will form a "tube" with boundary two parallel planes, which we glue together (so that we actually don't change the geometry). We can interpret this as having a copy of S^0 (which is a disjoint union of two points) for every point similar to how we have a copy of S^1 around every point for the $z-$axis and a copy of S^2 for the point, which we then identify opposite points. Notice that regardless of the dimensionality of the initial space which we were blowing up (a point, line/circle, or plane), the resulting blown up shape will always be $n - 1$ dimensional, and thus a hypersurface of sorts. This picture is also slightly misleading as it may make you believe there is a hole introduced into the space



(A) (B) (C)

Figure 6. Blowing up a point on the plane (A). Notice how it removes the self intersection. Blowing up the z-axis in $\mathbb{R}^3$ (B). Blowing up a point in $\mathbb{R}^3$ (C, top) and a line in a plane (C, bottom).

if you just look at the tube. However, there isn't an actual hole, it's just much harder to visualize if you actually identify the points (that being said, it does nontrivially alter the topology e.g. homology of the space). You may be angry that I'm being too handwavy and seem to be making up a random construction, but I promise you that behind all the claims that I have made so far, and all which are to come, have rigorous mathematical backing. The words are blow ups, or the proj construction, and these spaces are defined (at least in algebro-geometric settings) by algebraic equations.

5.3. **Final Stretch.** We will now blow up the copy of $\mathbb{P}^2$ (which we henceforth denote by Y) representing the double lines in $\mathbb{P}^5$ to obtain a new space which I'll call X. The resulting blown up hypersurface in X will be called E (it's formally called the "exceptional divisor"). Now, we want to look at the intersections of the degree six hypersurfaces which pass through the exceptional divisor and thus do not intersect each other at the Y.

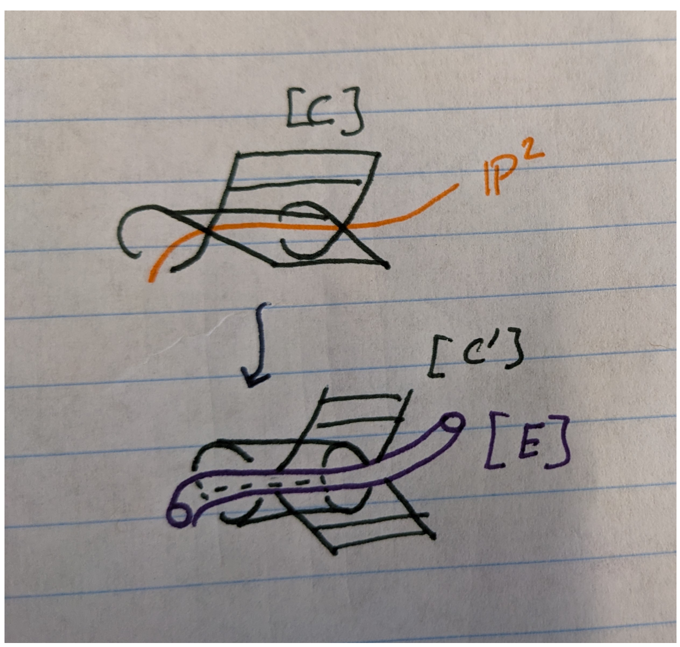


Figure 7. Blowing up the copy of $\mathbb{P}^2$ in $\mathbb{P}^5$.

In order to facilitate the computations, we will reformulate what we have done in terms of algebra. Formally, this setup is known as the Chow ring. We will write the degree 1 hypersurface in $\mathbb{P}^5$ which describes the conics which pass through some point as $[P]$ and the degree 2 hypersurfaces describing conics through a line as $[L]$. This may be a bit confusing, but these represent *any* such hypersurfaces of degree 1 and 2 in $\mathbb{P}^5$. We can define multiples of these symbols which correspond to disjoint unions of different copies of the hypersurfaces which they represent. For example, $3[P]$ would describe the union of 3 hyperplanes. Now we can take products of these things which correspond to their intersections, or more precisely, the intersection of generic representatives for these "classes". These work for arbitrary dimension, for example $[P]^2$ corresponds to the intersection of two generic hyperplanes i.e. a 3 dimensional plane. However, we will mainly consider products of 5 hypersurfaces as this will give us a finite number of points which we can count. We will denote by $[*]$ the class of a single point. Thus, we can reinterpret our prior results as $[P]^5 = [*]$, $[P]^4[L] = 2[*]$, or $[P]^3[L]^2 = 4[*]$. We also have the class $[C]$ on $\mathbb{P}^5$ which represents the degree 6 hypersurface dictating which conics are tangent to a given conic.

Now, we can consider these as classes on X as well by the following: we have a map $\varphi : X \to \mathbb{P}^5$ which comes from "crushing E to Y" and we can consider the preimage of a class in $\mathbb{P}^5$ under this map φ. What we are interested in is the hypersurfaces "passing through the portal", but if the hypersurface contains Y, the pre-image will contain E. We will denote the hypersurfaces passing through the portal with an extra prime. For example, $[P]$ does not intersect Y so the preimage of $[P]$ will just be $[P']$ while $[L]$ contains Y (as we discussed before, any double line poses as being tangent to any line i.e. any representative hypersurface for $[L]$ contains Y) and thus the preimage of $[L]$ will be $[L'] + [E]$ ($[E]$ is the class for E, it turns out to be "rigid" in some sense that it cannot really be deformed, so there is really just one object in this class). We will write this (perhaps a bit abusively, identifying $[L]$ with its preimage) as $[L] = [L'] + [E]$. Based on this notation, $[P']^2[L']^3 = 4[*]$ since looking at intersections in X removes the bad intersections at Y (but actually $[P]^2[L]^3 = 8[*]$, by Bézout).

What we want to compute is $[C']^5$, so what we need to analyze is the preimage of $[C]$ under φ. It turns out the correct expression is $[C] = [C'] + 2[E]$. The reason is that any representative for $[C]$ intersects Y twice. This is similar to how a line passes through the origin once (as $[L]$ contains Y once here), but $y^2 = x^3 - x^2$ passes through the origin twice, analogous to our scenario here. The way to see this is any line will intersect a conic at two points, and hence a double line will actually have two points of tangency with any conic. Now, if we had a nonsingular conic that had two points of tangency with a conic, there are two ways to deform this curve while staying tangent - keep either the first point tangent and vary the second point arbitrarily, or vice versa. This corresponds to moving along two different branches of a self intersection of this representative of $[C]$. Similarly, Y lies along a double self intersection of any such degree 6 hypersurface. Hence the preimage will actually contain 2 copies of $[E]$ (this is admittedly handwavy, but all of this can be rigorized).

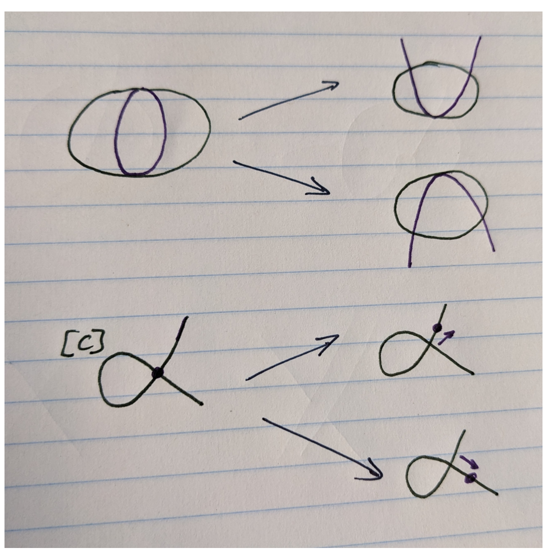


FIGURE 8. A conic which is tangent to another conic at two points (top) corresponds to a point that lies on a self intersection of the hypersurface $[C]$ (bottom).

Now, we want to compute $[C']^5 = ([C] - 2[E])^5$. We actually have all the tools to do this and the rest of the problem becomes purely algebraic! We noted earlier that two polynomial equations of the same degree could be linearly interpolated, and thus, regarding problems of intersection (which is what we care about), they can be regarded the same. Thus, $[L] = 2[P]$ and $[C] = 6[P]$. Thus, we have

$$[C']^5 = ([C] - 2[E])^5 = (6[P] - 2[E])^5.$$

Now to get rid of the $[E]$, we note that $[L'] + [E] = [L] = 2[P]$ and thus we have:

$$[C']^5 = (6[P] - 2[E])^5 = (6[P] - 2(2[P] - [L']))^5 = (2[P] + 2[L'])^5 = (2[P'] + 2[L'])^5$$

where we used $[P] = [P']$ in the last line. Now, we know all of the intersections $[P']^a[L']^b$, so we get:

$$\begin{aligned}
&(2[P'] + 2[L'])^5 \\
=&32([P']^5[L']^0 + 5[P']^4[L']^1 + 10[P']^3[L']^2 + 10[P']^2[L']^3 + 5[P']^1[L']^4 + [P']^0[L']^5) \\
=&32(1 + 5 * 2 + 10 * 4 + 10 * 4 + 5 * 2 + 1) = 3264
\end{aligned}$$

QED